Jessica Chen
HW 7

Output of FirstCode.scala:

```
jhc10024_nyu_edu@nyu-dataproc-m:~/hw7$ spark-shell --deploy-mode client -i FirstCode.scala
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/12 19:34:54 INFO SparkEnv: Registering MapOutputTracker
26/04/12 19:34:54 INFO SparkEnv: Registering BlockManagerMaster
26/04/12 19:34:54 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/04/12 19:34:54 INFO SparkEnv: Registering OutputCommitCoordinator
Spark context Web UI available at http://nyu-dataproc-m.c.hpc-dataproc-19b8.internal:39805
Spark context available as 'sc' (master = yarn, app id = application_1768446951072_10152).
Spark session available as 'spark'.
26/04/12 19:35:07 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
+----------------------+----------------------+--------------------+
|RESPONSE_TIME_mean_sec|DISPATCH_DELAY_mean_sec|TRAVEL_TIME_mean_sec|
+----------------------+----------------------+--------------------+
|1333.0495406571652    |406.7481000201368      |1066.8497603885698  |
+----------------------+----------------------+--------------------+

Median of RESPONSE_TIME: 1.0 seconds
Median of DISPATCH_DELAY: 3.0 seconds
Median of TRAVEL_TIME: 0.0 seconds
+------------------------+
|RESPONSE_TIME_stddev_sec|
+------------------------+
|3762.2326279659223      |
+------------------------+

Mode of BORO_NM:
+--------+-------+
|BORO_NM |count  |
+--------+-------+
|BROOKLYN|2094370|
+--------+-------+

Mode of RADIO_CODE:
+----------+------+
|RADIO_CODE|count |
+----------+------+
|75D       |709716|
+----------+------+

Mode of TYP_DESC_ENUM:
+-------------+------+
|TYP_DESC_ENUM|count |
+-------------+------+
|485          |709716|
+-------------+------+

Mode of CIP_JOBS:
+--------+-------+
|CIP_JOBS|count  |
+--------+-------+
|Non CIP |6365581|
+--------+-------+
```

Documentation:
FirstCode.scala Overview

This Spark application analyzes and cleans NYPD service call data stored in HDFS. It computes summary statistics, standardizes inconsistent formats, and produces a clean dataset for further analysis.

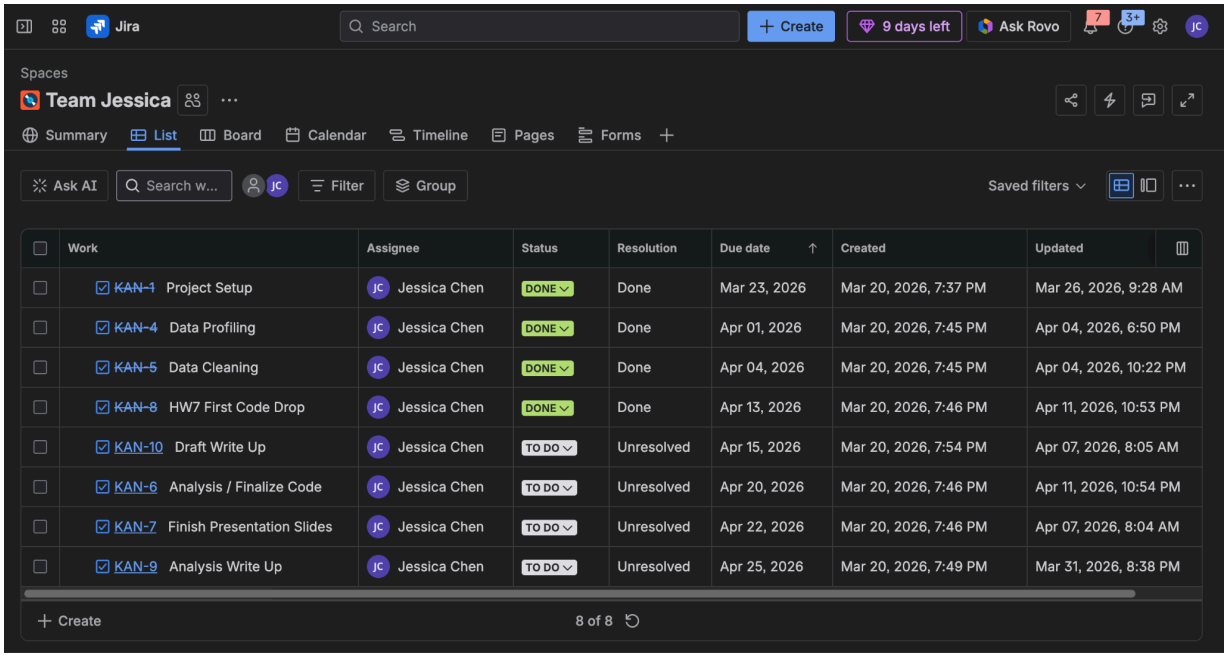
In Part 1, the code performs statistical analysis on three key numerical data columns: RESPONSE_TIME, DISPATCH_DELAY, and TRAVEL_TIME. These column values are first converted from HH:mm:ss format into total seconds so they can be analyzed numerically. The program then calculates the mean and median for each of these columns, and the standard deviation of RESPONSE_TIME to measure variability. For categorical variables such as BORO_NM, RADIO_CODE, TYP_DESC_ENUM, and CIP_JOBS, the mode (most frequent value) is determined using grouped counts.

In Part 2, the code focuses on cleaning and standardizing the data. Date fields like CREATE_DATE and INCIDENT_DATE are converted into proper date formats, while timestamp fields such as ADD_TS, DISP_TS, ARRIVD_TS, and CLOSNG_TS are formatted consistently. INCIDENT_TIME is also converted into a 12-hour AM/PM format. Additionally, categorical text fields are cleaned by trimming extra spaces and converting all values to uppercase to ensure consistency in analysis.

The dataset is read from HDFS at **hdfs:///user/jhc10024_nyu_edu/hw6/nypd_cfs_cleaned**. After processing, the cleaned data is saved back to HDFS at **hdfs:///user/jhc10024_nyu_edu/hw6/nypd_cfs_cleaned_2**.

The output is split into 50 partitions for better performance and saved in overwrite mode so it can be easily updated.

Jira Board:



Work	Assignee	Status	Resolution	Due date	Created	Updated
~~KAN-1~~ Project Setup	Jessica Chen	DONE	Done	Mar 23, 2026	Mar 20, 2026, 7:37 PM	Mar 26, 2026, 9:28 AM
~~KAN-4~~ Data Profiling	Jessica Chen	DONE	Done	Apr 01, 2026	Mar 20, 2026, 7:45 PM	Apr 04, 2026, 6:50 PM
~~KAN-5~~ Data Cleaning	Jessica Chen	DONE	Done	Apr 04, 2026	Mar 20, 2026, 7:45 PM	Apr 04, 2026, 10:22 PM
~~KAN-8~~ HW7 First Code Drop	Jessica Chen	DONE	Done	Apr 13, 2026	Mar 20, 2026, 7:46 PM	Apr 11, 2026, 10:53 PM
KAN-10 Draft Write Up	Jessica Chen	TO DO	Unresolved	Apr 15, 2026	Mar 20, 2026, 7:54 PM	Apr 07, 2026, 8:05 AM
KAN-6 Analysis / Finalize Code	Jessica Chen	TO DO	Unresolved	Apr 20, 2026	Mar 20, 2026, 7:46 PM	Apr 11, 2026, 10:54 PM
KAN-7 Finish Presentation Slides	Jessica Chen	TO DO	Unresolved	Apr 22, 2026	Mar 20, 2026, 7:46 PM	Apr 07, 2026, 8:04 AM
KAN-9 Analysis Write Up	Jessica Chen	TO DO	Unresolved	Apr 25, 2026	Mar 20, 2026, 7:49 PM	Mar 31, 2026, 8:38 PM